Lab 1: Intro to Networks, Unix, and Python Primer

> Name: Rachel (Younwoo) Park
> Student ID: 2172344
> Due: 10/5/2024 by 11:59pm

Unix Practice [20 points]

In this section you will learn or review some useful Unix commands. Test your commands and your Python program on a Linux machine.

1. Practice with directories and files

a. From a shell/ command prompt on either a linux device or a PC, what is the complete command to log in to the UCSC Unix Timeshare (unix.ucsc.edu)? Test your command by logging in and include a screenshot.

- `sudo ssh ypark55@unix.ucsc.edu`
-

```
base ~
sudo ssh ypark55@unix.ucsc.edu
Password:
Sorry, try again.
Password:
Sorry, try again.
Password:
The authenticity of host 'unix.ucsc.edu (128.114.104.57)' can't be established.
ED25519 key fingerprint is SHA256:nnvCsc0WN+RhcjTnJPy0AIzEFp6rSJb2kBXVH9NN0mo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'unix.ucsc.edu' (ED25519) to the list of known hosts.
ypark55@unix.ucsc.edu's password:
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file or directory
=========================================================================
*                                                                       *
*               Welcome to the Learning Technologies Timeshare!         *
*                                                                       *
*  Need help? Have a question? Something isn't working? We're here for you!  *
*  Email us at help@ucsc.edu, call us at 459-HELP (459-4357) or open a  *
*  support ticket at https://slughub.ucsc.edu                           *
*                                                                       *
*  Welcome to Fall Quarter, UCSC!  The big news this quarter is that our *
*  Linux lab is now running Mint, an Ubuntu variant!  Our new Kresge lab *
*  will also be running Mint when it opens later this quarter. The      *
*  timeshare is still running CentOS and will be migrating to AlmaLinux 8 *
*  during Winter break.  We are currently working on longer term plans for *
*  the timeshare and ways to evolve this service.  Stay tuned!          *
*                                                                       *
*      Have a wonderful Fall Quarter.  Stay safe and learn lots!  =)    *
*                                                                       *
=========================================================================

You are currently using 0% (22.0 KiB) of your 1.0 GiB quota.
-bash-4.2$
```

b. What command can be used to list all the files and directories (including hidden files) in the present working directory? Run the command in one of your directories and include a screenshot of the command and the resulting output. (If you don't have files, create a few random ones, you will need them later on).

- `ls -a`
-

```
ypark55@unix2.lt.ucsc.edu ~ (0.033s)
ls -a
.   .bash_history  .config  .esd_auth  .login    OldFiles     prelab1      welcome
..  .cache         .cshrc   .local     .viminfo  network_log  public_html
```

c. What command can be used to list all the files and directories in the present working directory in chronological order (based on the last modified date)? Run the command and Include a screenshot showing the resulting output.

- `ls -lt`
-

```
ypark55@unix2.lt.ucsc.edu ~ (0.104s)
ls -lt
total 9
drwxr-xr-x. 2 ypark55 users  2048 Oct  1 00:17 network_log
drwx--x--x. 8 ypark55 295716 2048 Oct  1 00:17 OldFiles
drwxr-xr-x. 2 ypark55 users  2048 Sep 30 23:59 prelab1
drwxr-xr-x. 2 ypark55 295716 2048 Apr 15 21:14 public_html
-rw-------. 1 ypark55 295716  224 May  7  2014 welcome
```

- if you want to see the hidden files, you can use `ls -alt`

```
ypark55@unix2.lt.ucsc.edu ~ (0.07s)
ls -alt
total 272
-rw-------. 1 ypark55 users     915 Oct  1 17:58 .bash_history
drwx--x--x. 8 ypark55 295716   2048 Oct  1 17:57 .
drwxr-xr-x. 4 ypark55 users    2048 Oct  1 17:57 .config
-rw-------. 1 ypark55 users      16 Oct  1 17:57 .esd_auth
drwxrwxrwx. 2 root    root   253952 Oct  1 07:40 ..
drwxr-xr-x. 2 ypark55 users    2048 Oct  1 00:17 network_log
drwx--x--x. 8 ypark55 295716   2048 Oct  1 00:17 OldFiles
drwxr-xr-x. 2 ypark55 users    2048 Sep 30 23:59 prelab1
-rw-------. 1 ypark55 users    1358 Sep 30 23:48 .viminfo
drwxr-xr-x. 3 ypark55 users    2048 Sep 30 22:38 .cache
drwxr-xr-x. 3 ypark55 users    2048 Sep 30 22:38 .local
drwxr-xr-x. 2 ypark55 295716   2048 Apr 15 21:14 public_html
-rw-------. 1 ypark55 295716    224 May  7  2014 welcome
-rw-------. 1 ypark55 295716    742 May  7  2014 .login
-rw-------. 1 ypark55 295716   1613 May  7  2014 .cshrc
```

d. What command is used to display the current directory?

- `pwd`
- output: `/afs/cats.ucsc.edu/users/u/ypark55`

e. What command is used to create a directory named "prelab 1" in your home directory? What is the complete path name to this directory?

- `mkdir ~/prelab1`
- path: `/afs/cats.ucsc.edu/users/u/ypark55/prelab1`

f. What command is used to move to the root directory?

- `cd /`

2. Working with Files

Copy the two files file1.txt and file2.txt into your directory and use them for questions (a),(b) and (c)

a. What is the single line command to display the contents of the two files?

- `cat file1.txt file2.txt`

b. What complete command allows you to see the line by line differences between these two files? Run the command and include a screenshot showing the results of running the command.

- `diff file1.txt file2.txt`

```
ypark55@unix2.lt.ucsc.edu ~/prelab1 (0.028s)
diff file1.txt file2.txt
1c1,3
< CSE 150! Embark on your journey into Linux with hands-on command line exercises.
---
> Success! You've now become adept at Linux file operations using the command line. Keep
up the great work!
>
> Don’t forget to vote!
```

c. What complete command is used to concatenate these two files and store the output to a new file named file3.txt? Run the command and Include screenshot showing the command and the resulting contents of file3.txt

```
cat file1.txt file2.txt > file3.txt
cat file3.txt
```

```
ypark55@unix2.lt.ucsc.edu ~/prelab1 (0.04s)
cat file1.txt file2.txt > file3.txt

ypark55@unix2.lt.ucsc.edu ~/prelab1 (0.027s)
cat file3.txt
CSE 150! Embark on your journey into Linux with hands-on command line exercises.

Success! You've now become adept at Linux file operations using the command line. Keep up
 the great work!

Don’t forget to vote!
```

d. What complete command would you use to move file3. txt from your current directory to a directory having relative path "/dir2"?

- `mv file3.txt ../dir2/`

e. What complete command would you use to delete a file named file 1 .txt? Can the file be recovered once this command is issued? Why or why not? Explain.

- `rm file1.txt`
- After using the `rm` command, it cannot be recovered. Although, the file data may still exist on the disk until it is overwritten. (If you know the inode number, it may be possible to recover the file.)

f. What command would you use to search for the string "vote" in "file.txt" in the same directory as "file3.txt" is located in?

- `grep "vote" file3.txt`

3. Moving files around the network

Create a directory named “network_log” (as shown below) in your home directory on the Unix Timeshare Server. In that directory, create the structure below with three .txt files, a png and a pdf file.

```
<user_directory>
└── network_log
├── f1.txt
├── f2.txt
├── f3.txt
├── a.png
└── b.pdf
```

What is the complete command to non-recursively copy all of the .txt files into your local computer’s current working directory? (This can either be on your own laptop or the lab computer.) Using this command, demonstrate that your command works by displaying a screenshot of the directory listing on the Unix server and then after the copy, show the directory listing on your local machine.

```
scp ypark55@unix.ucsc.edu:/afs/cats.ucsc.edu/users/u/ypark55/network_log/\*.txt .
```

Unix server:

- ```
  ypark55@unix2.lt.ucsc.edu ~/prelab1 (0.183s)
  cd ../network_log/

  ypark55@unix2.lt.ucsc.edu ~/network_log (0.031s)
  ls
  a.png  b.pdf  f1.txt  f2.txt  f3.txt
  ```

Local machine:

- ```
  base ~ (4.098s)
  scp ypark55@unix.ucsc.edu:/afs/cats.ucsc.edu/users/u/ypark55/network_log/\*.txt .

  The Unix timeshare uses the BLUE password, which is different than the
  password you use for most other systems. If you need to reset your BLUE
  password, head over to  https://cruzid.ucsc.edu

  ypark55@unix.ucsc.edu's password:

  base ~ (0.184s)
  lsd
    Applications              Desktop      f1.txt    'Google Drive'    Movies      Postman
    'Creative Cloud Files'    Documents    f2.txt    karabiner         Music       project
    cse_140                   Downloads    f3.txt    Library           Pictures    Public
  ```

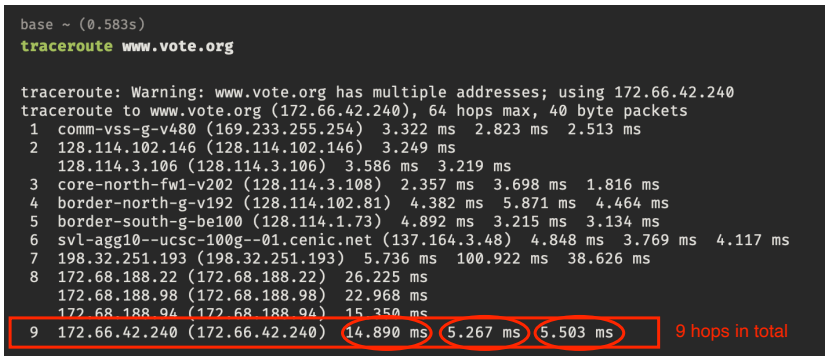
Traceroute

The Traceroute program sends a series of probes (messages carried in a packet) from a source to a destination over the network. Each probe travels part-way to the destination until finally the destination is reached. The probes travel “hop by hop”, meaning from one router to the next on the path to the destination. In each iteration the probes incrementally get one hop closer to the destination until finally, in the last iteration, the destination is reached.

Note: When working on the problems, if Traceroute is hanging and doesn’t complete, talk to your TA. Multiple lines with _ **and no completion is not what we are looking for in this exercise and will not receive credit.** _ most likely means the router receiving the probe is not responding. Overall the majority of lines must display a normal response from the router

4. Basic traceroute

Run Traceroute between your computer and www.vote.org. Include a screenshot of the output and answer the following questions based on the output in your screenshot:



```
base ~ (0.583s)
traceroute www.vote.org

traceroute: Warning: www.vote.org has multiple addresses; using 172.66.42.240
traceroute to www.vote.org (172.66.42.240), 64 hops max, 40 byte packets
 1  comm-vss-g-v480 (169.233.255.254)  3.322 ms  2.823 ms  2.513 ms
 2  128.114.102.146 (128.114.102.146)  3.249 ms
    128.114.3.106 (128.114.3.106)  3.586 ms  3.219 ms
 3  core-north-fw1-v202 (128.114.3.108)  2.357 ms  3.698 ms  1.816 ms
 4  border-north-g-v192 (128.114.102.81)  4.382 ms  5.871 ms  4.464 ms
 5  border-south-g-be100 (128.114.1.73)  4.892 ms  3.215 ms  3.134 ms
 6  svl-agg10--ucsc-100g--01.cenic.net (137.164.3.48)  4.848 ms  3.769 ms  4.117 ms
 7  198.32.251.193 (198.32.251.193)  5.736 ms  100.922 ms  38.626 ms
 8  172.68.188.22 (172.68.188.22)  26.225 ms
    172.68.188.98 (172.68.188.98)  22.968 ms
    172.68.188.94 (172.68.188.94)  15.350 ms
 9  172.66.42.240 (172.66.42.240)  14.890 ms  5.267 ms  5.503 ms
```


a. What does each of the numbered lines represent?

- Each numbered line represents a hop
  - A hop is move that data makes as it goes from one network device/router to another
  - Each hope shows the network latency between the source and the destination.
  - The first hop is the source, and the last hop is the destination.

b. Choose one of the output rows and circle the 3 time measurements displayed. Explain what they represent.

- The three time measurements represent the length of time (in ms) to send the ICMP(Internet Control Message Protocol) packet from the source to the destination and back to the source.

c. How do you expect the time measurements in row #1 to differ from row #2? Greater? Less? The same? Explain your reasoning by focusing on the operation of each row in the program output

- It is expected that time measurements in row #1 is **less** than row #2 because the first hop is typically the local router which is close to the source. As the packets move further from the source, the latency tends to increase due to greater physical distance and other factors. (row #1 time < row #2 time)

d. Define hop count. Based on your definition, how many hops are there between your computer and the final destination? Mark up your screenshot as needed

- Hop count refers to the number of network devices that a packet travels until in reaches its destination. There are 9 hops in this traceroute output.

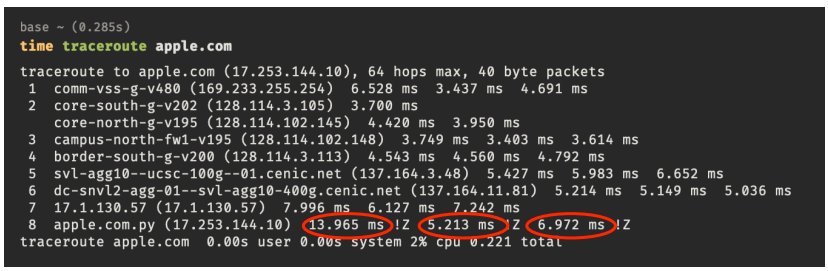
source

5. North America

Run Traceroute between your computer and a destination on this continent (North America) at three different run times: morning, mid-day and night. Make sure that you are in the SAME location - i.e. at home or at school etc. for each of the runs. Record your results in the table below capturing the time to reach the final destination.

Destination Hostname: apple.com					**Location**: A tool such as IP Location Finder can be used to determine the approximate geographic location of a host.	
		Time to reach final destination				
Run #	Execution Time	RTT 1	RTT 2	RTT 3	Average RTT to final destination	# hops
1 (morning)	0.221	13.965	5.213	6.972	8.717	8
2 (mid-day)	5.328	4.897	11.650	8.673	8.407	8
3 (night)	10.354	11.575	11.041	12.173	11.596	8

a. Include a screenshot of Run #1 (only) and circle all data used to calculate the average RTT to the destination host.



```
base ~ (0.285s)
time traceroute apple.com
traceroute to apple.com (17.253.144.10), 64 hops max, 40 byte packets
 1  comm-vss-g-v480 (169.233.255.254)  6.528 ms  3.437 ms  4.691 ms
 2  core-south-g-v202 (128.114.3.105)  3.700 ms
    core-north-g-v195 (128.114.102.145)  4.420 ms  3.950 ms
 3  campus-north-fw1-v195 (128.114.102.148)  3.749 ms  3.403 ms  3.614 ms
 4  border-south-g-v200 (128.114.3.113)  4.543 ms  4.560 ms  4.792 ms
 5  svl-agg10--ucsc-100g--01.cenic.net (137.164.3.48)  5.427 ms  5.983 ms  6.652 ms
 6  dc-snvl2-agg-01--svl-agg10-400g.cenic.net (137.164.11.81)  5.214 ms  5.149 ms  5.036 ms
 7  17.1.130.57 (17.1.130.57)  7.996 ms  6.127 ms  7.242 ms
 8  apple.com.py (17.253.144.10)  13.965 ms !Z  5.213 ms !Z  6.972 ms !Z
traceroute apple.com  0.00s user 0.00s system 2% cpu 0.221 total
```

b. Did the number of routers in the path change during your 3 different runs? If so, why do you think it changed? If not, why do you think it didn't?

- The number of routers in the path did not change during the 3 different runs. They all had 8 hops.
- I think the number of routers did not change because of stable network conditions. Because the network path is stable, the routers and routes would not change unless there is an issue like congestion or link failure. There was likely no congestion or failures, leading to the consistent number of routers.

6. Different Continents

Repeat process to a destination on a different continent.

Destination Hostname:	117.52.4.46					
Location of host:	Seoul, Korea					
		Time to reach final destination				
Run #	Execution Time	RTT 1	RTT 2	RTT 3	Average RTT to final destination	# hops
1 (morning)	42.476	152.256	145.441	144.201	147.299	14
2 (mid-day)	41.021	152.015	152.266	148.115	150.799	14
3 (night)	36.137	137.376	146.979	148.465	144.273	15

a. Include a screenshot of Run #1 (only) and circle all data used to calculate the average RTT to the destination host.

```
base ~ (42.54s)
time traceroute 117.52.4.46
traceroute to 117.52.4.46 (117.52.4.46), 64 hops max, 40 byte packets
 1  comm-vss-g-v480 (169.233.255.254)  167.053 ms  6.068 ms  5.667 ms
 2  128.114.102.146 (128.114.102.146)  5.782 ms
    128.114.3.106 (128.114.3.106)  6.298 ms  6.023 ms
 3  campus-north-fw1-v195 (128.114.102.148)  4.127 ms  6.856 ms  3.968 ms
 4  border-north-g-v192 (128.114.102.81)  5.912 ms  5.596 ms  4.682 ms
 5  dc--lax-agg10--ucsc--100g.cenic.net (137.164.23.182)  289.639 ms  13.316 ms  12.372 ms
 6  * * *
 7  1.213.115.145 (1.213.115.145)  14.849 ms  12.683 ms  11.545 ms
 8  203.233.12.133 (203.233.12.133)  164.814 ms
    1.208.148.69 (1.208.148.69)  157.581 ms
    1.208.105.17 (1.208.105.17)  162.679 ms
 9  * * *
10  1.208.164.14 (1.208.164.14)  150.009 ms
    61.42.234.154 (61.42.234.154)  154.849 ms
    1.208.113.106 (1.208.113.106)  153.530 ms
11  1.208.165.170 (1.208.165.170)  155.223 ms
    1.208.164.166 (1.208.164.166)  159.606 ms *
12  1.208.167.118 (1.208.167.118)  154.889 ms
    61-111-61-150.kidc.net (61.111.61.150)  162.346 ms
    61-111-40-234.kidc.net (61.111.40.234)  148.405 ms
13  61-111-40-62.kidc.net (61.111.40.62)  146.974 ms  138.660 ms
    182.162.36.226 (182.162.36.226)  144.854 ms
14  117.52.4.46 (117.52.4.46)  152.256 ms  145.441 ms  144.201 ms
traceroute 117.52.4.46  0.01s user 0.02s system 0% cpu 42.476 total
```

b. Try to identify an overseas link with increased delay and mark/label it in your screenshot.

- Hop 5 is the overseas link with increased delay. There is a significant increase in the latency at hop 5 compared to the other hops.

7. General questions about your experiment

a. Compare the average intra-continent and inter-continent results. Are your results what you would expect? Explain

- The results were as expected. The intra-continent traceroute has lower RTT compared to the inter-continent traceroute.Since there is a greater physical distance between the source and destination, the inter-continent traceroute has higher RTT.

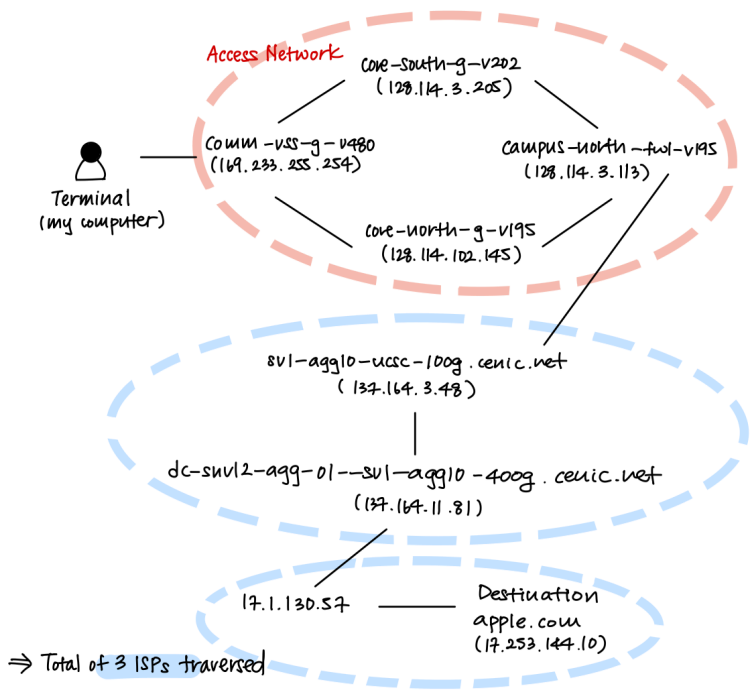
b. For a given source and destination, find a row in your table for which the reported RTT values are different. What could account for the difference in delay?

- When looking at the reported RTT values of the three runs of the destination 117.52.4.46 (seoul, korea), the RTT for mid-day is slightly higher compared to the others. There might have been more traffic during the mid-day run, causing congestion and increased latency. Furthermore, while Korea itself may have been experiencing lower traffic (because it is night time), the backbone networks may still be handling significant global traffic, impacting the RTT.

8. Draw the North America network path

Think about our “Model Network on Paper” from the lecture and consider the traceroute output in Question 5. Make a drawing of the network. Make sure your drawing uses Nodes and Edges (routers and links) and include your computer as the starting Terminal.

- Identify your Access Network – mark and label it on your drawing. (An Access Network is the network used to connect to the Internet.)
- Routers with similar names and/or similar IP addresses can be considered as part of the same ISP. Put a dotted circle around any such groupings of nodes in your drawing to indicate these networks.
- Try to identify the number of ISPs that your packets traversed from the source to destination and write this number on the drawing



Practice with Python

References: Refer to the Python tutorial above and to the previous section for Traceroute.

9. Practice with Python

Every time you surf the web, your browser displays a web document that is hosted on a web server. In order to display the web document, a request message must first be sent from your computer to the web server. Then, when a response message is returned and received by the browser, the requested page is displayed. Sometimes the messages must be forwarded by many routers on the path from your computer to the web server. In this exercise we will use traceroute to look at the “length” of such paths.

Write a simple utility program that accepts a web server name as input and then invokes the traceroute program to determine the number of hops from the source (where the program is invoked) to the web server.

Program operation:

- prompt the user for a host name (a web server)
- Hard coding is not allowed:
  - read the host name from the command line
  - print the output to stdout
- invoke the ‘traceroute’ command to determine the number of hops to a web server. Allow Traceroute to run for 20 seconds before checking its output

Results

```
base ~ (12.551s)
python -u "/Users/rachelie/Library/CloudStorage/GoogleDrive-rachelie.dwg@gmail.com/My Drive/sk
ku/6학 기 /[CSE 150] Computer Networks /labs/ypark55-lab1.py"
Enter a server name: www.abcnews.com
9 hops to www.abcnews.com

base ~ (5.233s)
python -u "/Users/rachelie/Library/CloudStorage/GoogleDrive-rachelie.dwg@gmail.com/My Drive/sk
ku/6학 기 /[CSE 150] Computer Networks /labs/ypark55-lab1.py"
Enter a server name: unreachable
Traceroute: failed

base ~ (24.987s)
python -u "/Users/rachelie/Library/CloudStorage/GoogleDrive-rachelie.dwg@gmail.com/My Drive/sk
ku/6학 기 /[CSE 150] Computer Networks /labs/ypark55-lab1.py"
Enter a server name: 117.52.4.46
Traceroute: failed
```

- In my program, when the host could be reached within 20 seconds, the program prints the number of hops.
- If the host does not exist or is unreachable within 20 seconds, the program prints `Traceroute: failed`
- The program had the same output as the expected output.